

GOOD FRUIT GROWER

NEW TECHNOLOGY / BUYERS' GUIDE

GOOD FRUIT
GROWER

The essential resource for growers



Speeding up computer vision systems is an understatement when describing the latest tools available to growers looking for quick crop data, such as this Green Atlas Cartographer zipping through a Yakima Valley, Washington, cherry block in April. Learn more about this and other systems on Page 36.

TJ HULL/NAXIS/GOOD FRUIT GROWER

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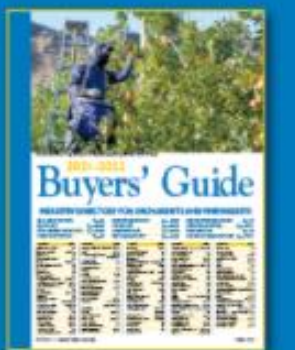
Can a "project management" approach to specialty crop producers' need for automation accelerate the pace of commercialization?

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Washington State University graduate student Chenchen Kang sets up a new hyperspectral sensor system as part of the Smart Irrigation project in Prosser in July.

Can hyperspectral cameras usher in a new era of smart irrigation for wine grape growers?

by Kate Prengaman

Growers know water stress when they see it: Shoot tips stop their rapid growth, tendrils turn limp and petioles droop, and leaves begin to bend.

But hyperspectral cameras and multispectral cameras can see signs of water stress long before it's apparent to even the well-trained eye.

Using such sensors to estimate plant water stress directly, rather than indirectly with soil moisture sensors or weather forecasts and evapotranspiration data, could usher in a new era of precision irrigation, according to Markus Keller, a viticulture professor at Washington State University. In the arid West, irrigation remains one of growers' most important tools to control fruit quality, he said, and is often considered part of the art of wine grape growing.

"For many growers it's not as much an art as a black box," he said. "It's very difficult to manage things properly if we can't measure it."

Keller and his graduate student, Geraldine Diverres, teamed up with engineers from WSU's Center for



The human eye can only see wavelengths of light in the 300 to 730 nanometer range, but hyperspectral cameras can record wavelengths across a wider spectrum, including ultraviolet and infrared. Washington State University researchers found that the patterns of light reflecting off grapevines can offer insight into vine water stress.

Precision and Automated Agriculture Systems (CPAAS) to see how hyperspectral sensing technology, which picks up far more wavelengths than the human eye, could guide irrigation. The team received nearly \$700,000 in 2018 from the National Science Foundation and the U.S. Department of Agriculture to support the work.

They aim to develop a reliable, robust and repeatable model to estimate vine water status, said CPAAS professor Manoj Karkee, the principal investigator on

the project, along with CPAAS director Qin Zhang and computer science professor Yinghui Wu of Case Western Reserve University.

For the first step, the new hyperspectral sensors must be correlated with traditional measurements of plant water stress: stomatal conductance and leaf water potential.

If stomata are the windows in the leaves for gas exchange, stomatal conductance refers to how open or closed the windows are, Diverres said. More open, they lose more water. Vines have to find this sweet spot between photosynthesis and evapotranspiration and water stress.

In early July, she taught another WSU graduate student, Priyanka Upadhyaya, how to use a tool called a gas exchange analyzer to measure the stomatal conductance of experimental vines facing different levels of water stress. Diverres also used a pressure bomb to measure the leaf water potential of the same leaves Upadhyaya sampled, while Chenchen Kang, an engineering grad student, used the camera sensors to scan the same vines.

Karkee said the light reflectance recorded by the camera does correlate with leaf water potential, and the model they've built so far can predict plant water status within 80 percent accuracy.

Combining all that data, along with soil moisture data, will help to correlate what leaf spectral signatures indicate with respect to vine water stress.

Next, Wu, the computer scientist on the project, will

build this information into one model to predict how much water vines need for the upcoming week, turning it from a water stress measure to an irrigation management tool.

Then, that model will need to be tested.

"In the next year or two, we are hoping to have a prototype system for decision making," Karkee said. "It's a framework for continual improvement year after year."

Eventually, a drone or ATV carrying a hyperspectral camera should be able to map the vine water need variability across the block, Keller said.

"One of the main problems when you farm is variability," he said. "When you irrigate with a one-size-fits-all approach, some vines will be overwatered, some will be underwatered, and some may be just right."

Currently, growers have few tools for variable rate irrigation, Keller said, but he believes it's coming.

For example, Nick Dokoozlian, vice president of research at E. & J. Gallo, recently shared insights from a precision irrigation trial he led at a precision viticulture symposium, breaking a vineyard into different irrigation zones due to soil and water need variation.

That approach isn't yet practical for commercialization, said Russ Smithyman, the chair of the National Grape Research Alliance, which hosted the virtual symposium where Dokoozlian spoke. But the tools Keller and Karkee are developing will have value for growers either looking to manage their farms to be more uniform or to better understand and use the variability, he said.

Keller and Karkee agreed that the technology to deploy vine-by-vine irrigation isn't yet available, but it's coming.

"It's expensive. But at one point drip was too expensive to make it in the marketplace, and now everyone uses drip," Keller said. "This is a project that's ahead of its time." ●



Washington State University graduate student Geraldine Diverres collects leaf water potential data with an old tool, the pressure bomb, to understand the relationship between those measurements and how a new hyperspectral sensor can see signs of water stress.

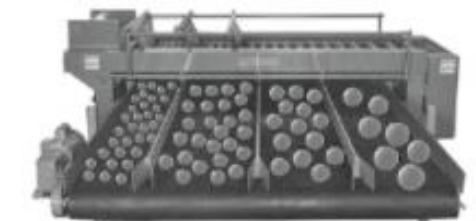


Using this gas exchange analyzer, WSU grad student Priyanka Upadhyaya gathers data on stomatal conduction, as part of the Smart Irrigation project.



Kang is setting up a new hyperspectral sensor for 2021 field work. The setup includes two other cameras — one thermal camera and one 3D camera — that help to understand canopy depth. Eventually, a hyperspectral sensor could be flown on a drone or mounted to an ATV to provide data on irrigation needs.

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